

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 3 – SHORT ANSWER TEST**

|                         |                                                                                                              |                      |                                       |
|-------------------------|--------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------|
| <b>Course Code:</b>     | CSA0309                                                                                                      | <b>Course Title:</b> | Data Structures                       |
| <b>CO Assessed:</b>     | CO5 – Manipulation (BL3, BL4)                                                                                | <b>Assessment:</b>   | Assessment Tool 3 – Short Answer Test |
| <b>Weightage:</b>       | 30% of CIA                                                                                                   | <b>Total Marks:</b>  | 100                                   |
| <b>CO2 Statement:</b>   | Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem. |                      |                                       |
| <b>Student Name:</b>    | Bankar. R                                                                                                    | <b>Reg. No.:</b>     | 192525131                             |
| <b>Section / Batch:</b> | Slot - B                                                                                                     | <b>Date:</b>         | 02/09/2026                            |

**Code of Conduct**

I Bankar. R [192525131] (Name / Reg No)  
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Instructions**

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

| Section / Topic         | Focus                         | Q Nos | Marks            |
|-------------------------|-------------------------------|-------|------------------|
| Minimum Spanning Tree   | Kruskal's Algorithm           | 1     | 20               |
| Graph Sorting           | Topological Sort              | 2     | 20               |
| Shortest Path Algorithm | MST & Dijkstra's Algorithm    | 3     | 20               |
| Shortest Path Algorithm | Dijkstra's Algorithm          | 4     | 20               |
| Minimum Spanning Tree   | Prim's or Kruskal's Algorithm | 5     | 20               |
| <b>GRAND TOTAL:</b>     |                               |       | <b>100 Marks</b> |

# Annexure A - Short Answer Test

- Perform the following operations in the Singly Linked List for the following input.  
Input: 22->77->55->88->  
i. Insert a new-node 110 at the end of the list.  
ii. Insert a new-node 101 at the position 4 of the list.  
iii. Delete a node at the beginning of the single linked list.  
iv. Delete a node at the position 3 of the single linked list.
- Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10 ( ), 4) Push (90), 5) Push (36), 6) Push (11)
- Convert the following infix into a postfix expression.  $\rightarrow B - (C/D + (E \% F * G) / H) * J$
- Evaluate the given postfix expression. 53+62/\*35\*+
- Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
- Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115.
- Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19. At each stage of heap tree construction, ensure heap properties are satisfied.
- Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), insert (3), delete (9), delete (60), insert (3), search (12).
- Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?
- Consider the following graph & construct BFS.



Graph with six nodes

## Rubrics for Calculation

| Criteria                     | Weightage | Level 4 - Exemplary                                                    | Level 3 - Proficient               | Level 2 - Developing           | Level 1 - Beginning           |
|------------------------------|-----------|------------------------------------------------------------------------|------------------------------------|--------------------------------|-------------------------------|
| Conceptual Understanding     | 30        | Demonstrates complete and accurate understanding of concepts           | Good understanding with minor gaps | Basic understanding            | Limited understanding         |
| Accuracy of Answer           | 25        | Answers are completely correct and relevant                            | Mostly correct with minor errors   | Partially correct              | Mostly incorrect              |
| Problem-Solving & Reasoning  | 20        | Provides logical reasoning and appropriate steps                       | Good reasoning with minor gaps     | Basic reasoning                | Lacks logical reasoning       |
| Clarity & Relevance          | 15        | Answers are precise, clear, concise, and directly address the question | Mostly clear and relevant          | Somewhat unclear or incomplete | Irrelevant or unclear answers |
| Time Management & Completion | 10        | Completes all questions accurately within the time                     | Completes most questions on time   | Requires additional time       | Unable to complete the test   |

| SFaculty In-charge | Course Coordinator | Program Director |
|--------------------|--------------------|------------------|
| Signature:         | Signature: _____   | Signature: _____ |
| Date: _____        | Date: _____        | Date: _____      |

Input: 22 77 55 88  
Singly linked list:

i.) Insert 110 at end:

22->77->55

ii.) Insert 101 at Pos 4

22->77->

iii.) Delete 88

iv.) Delete

2.) So

i) Input: 22 77 55 88  
Singly linked list:

ii) Insert 110 at end:

22 → 77 → 55 → 88 → 110

iii) Insert 101 at Position 4:

22 → 77 → 55 → 101 → 88 → 110

iv) Delete Beginning

77 → 55 → 101 → 88 → 110

v) Delete Position 3:

77 → 55 → 88 → 110

2) Stack operation:

Operation: Push(90), Push(36), Push(11), Push(88), Pop(), Pop()

Principle: LIFO (Last In First out)

| Operation | Stack (Bottom → Top) | Top |
|-----------|----------------------|-----|
|           |                      | 90  |
| Push(90)  | 90                   |     |
| Push(36)  | 90, 36               | 36  |
| Push(11)  | 90, 36, 11           | 11  |
| Push(88)  | 90, 36, 11, 88       | 88  |
| Pop()     | 90, 36, 11           | 11  |
| Pop()     | 90, 36               | 36  |

3. Infix to Postfix:

$$\text{Given: } B - (C/D + E * F * G) / H)^* J$$

Steps:

$$C/D \rightarrow CD/$$

$$E * F \rightarrow EF*$$

$$(E * F) * G \rightarrow EF * G *$$

$$(E * F * G) / H \rightarrow EF * G * H /$$

$$C/D + \dots \rightarrow CD / EF * G * H / +$$

$$\text{Multiply by } J \rightarrow CD / EF * G * H / + J *$$

Subtract from B

Postfix:

$$B CD / EF * G * H / + J * -$$

4.) Evaluate Postfix Expression

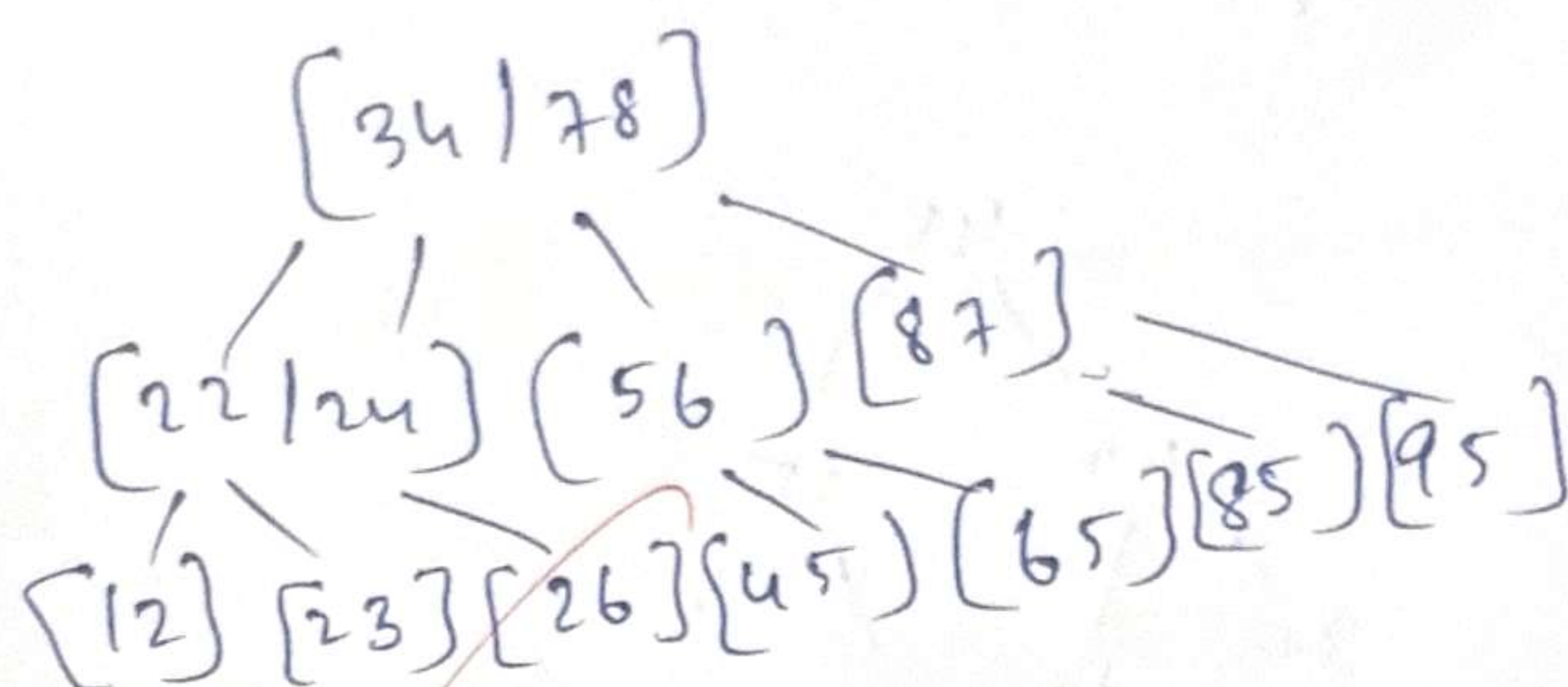
$$\text{Given: } 53 + 62 / * 35 * +$$

Postfix Expression: 39

| Token | Operation | Stack    |
|-------|-----------|----------|
| 5     | Push      | 5        |
| 3     | Push      | 5, 3     |
| +     | 5 + 3     | 8        |
| 6     | Push      | 8, 6     |
| 2     | Push      | 8, 6, 2  |
| /     | 6 / 2     | 8, 3     |
| *     | 8 * 3     | 24       |
| 3     | Push      | 24, 3    |
| 5     | Push      | 24, 3, 5 |
| *     | 3 * 5     | 24, 15   |
| +     | 24 + 15   | 39       |

B-Tree of order 3

Elements : 34, 26, 45, 12, 87, 56, 78, 22, 24, 26, 65, 85, 95



Final Structure :

\* Root : 34, 78

\* Second level : 22, 24, [56 | 87]

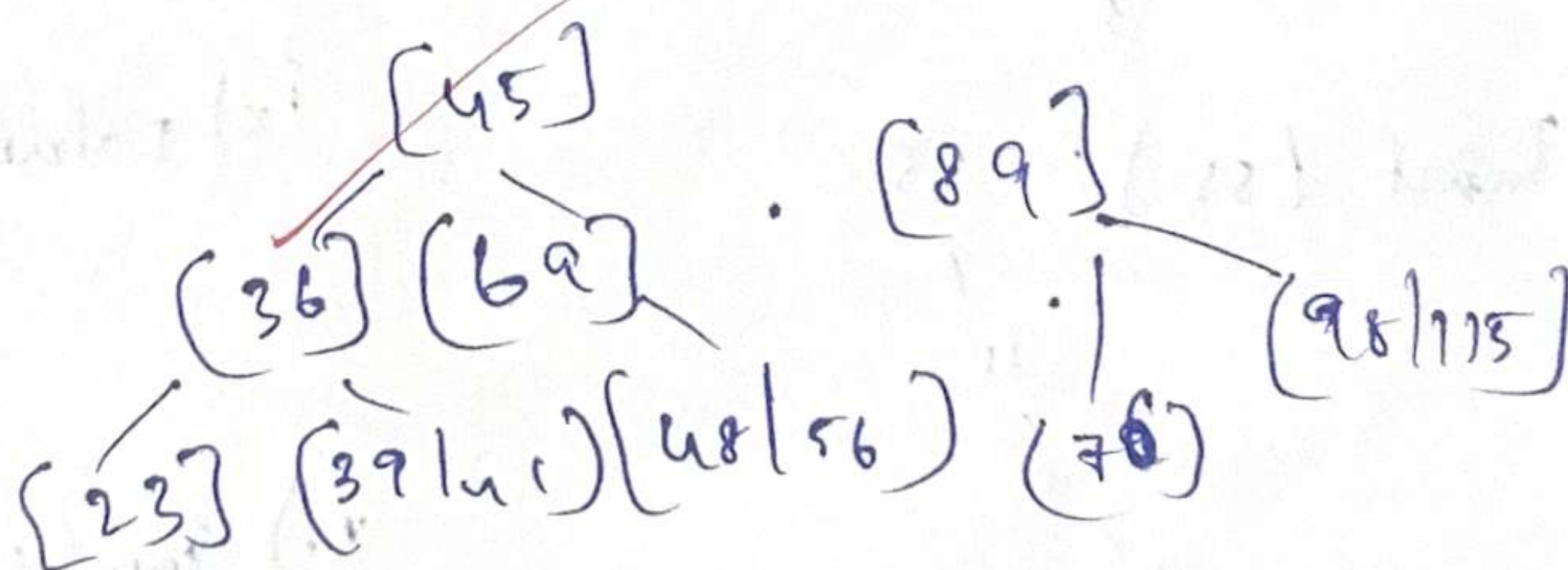
\* Leaf level : 12 | 23 | 26 | 45 | 65 | 85 | 95

2-3-4 Tree

Elements :

45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48

Final 2-3-4 Tree



\* Root : 45

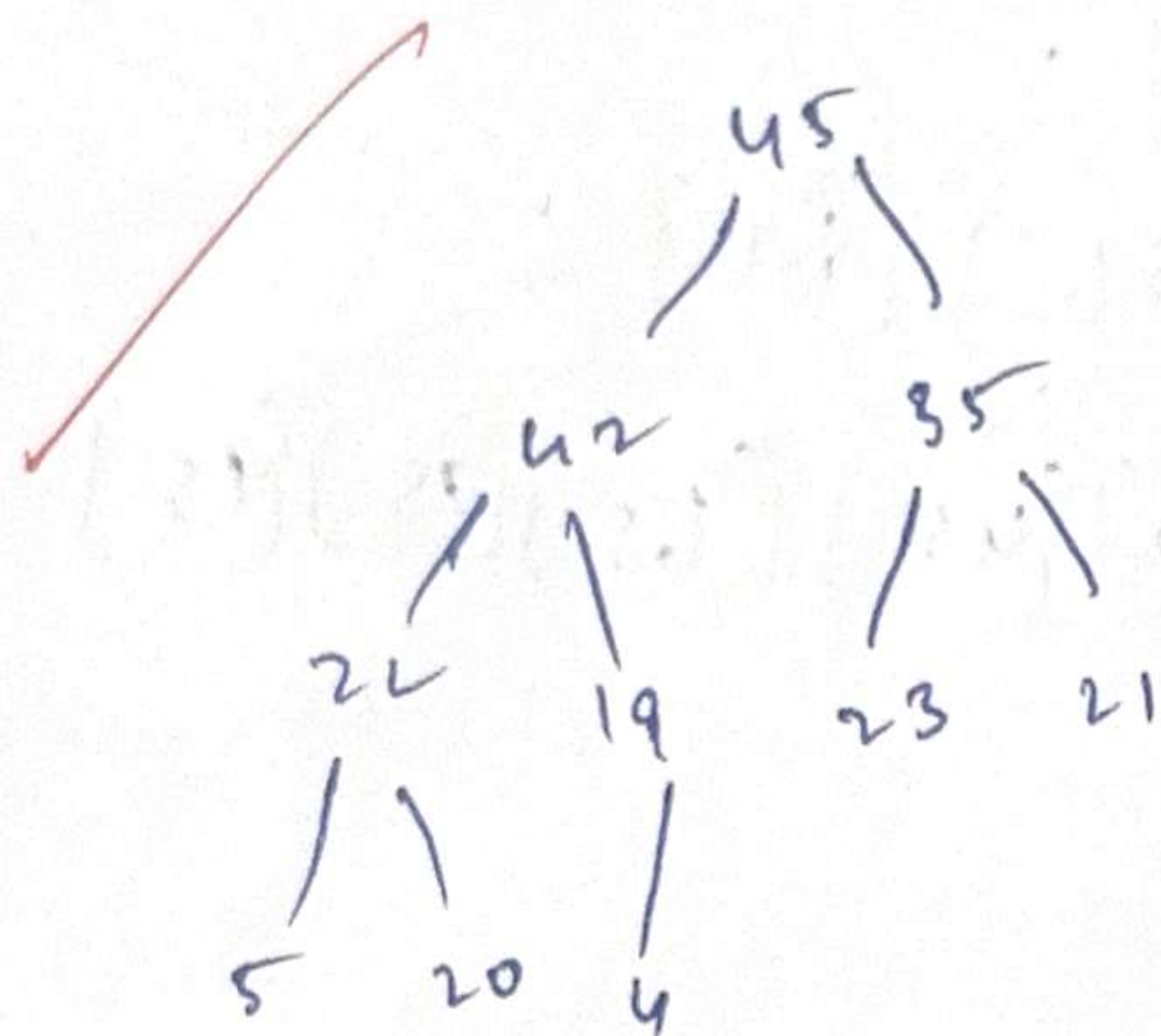
\* Second level : 36, 69 | 89

\* Leaf level : 23, 39, 41, 48, 56, 76, 98, 115

7.) Max heap:

Elements: 20, 35, 23, 22, 4, 45, 21, 5, 19, 14

Find Max heap:

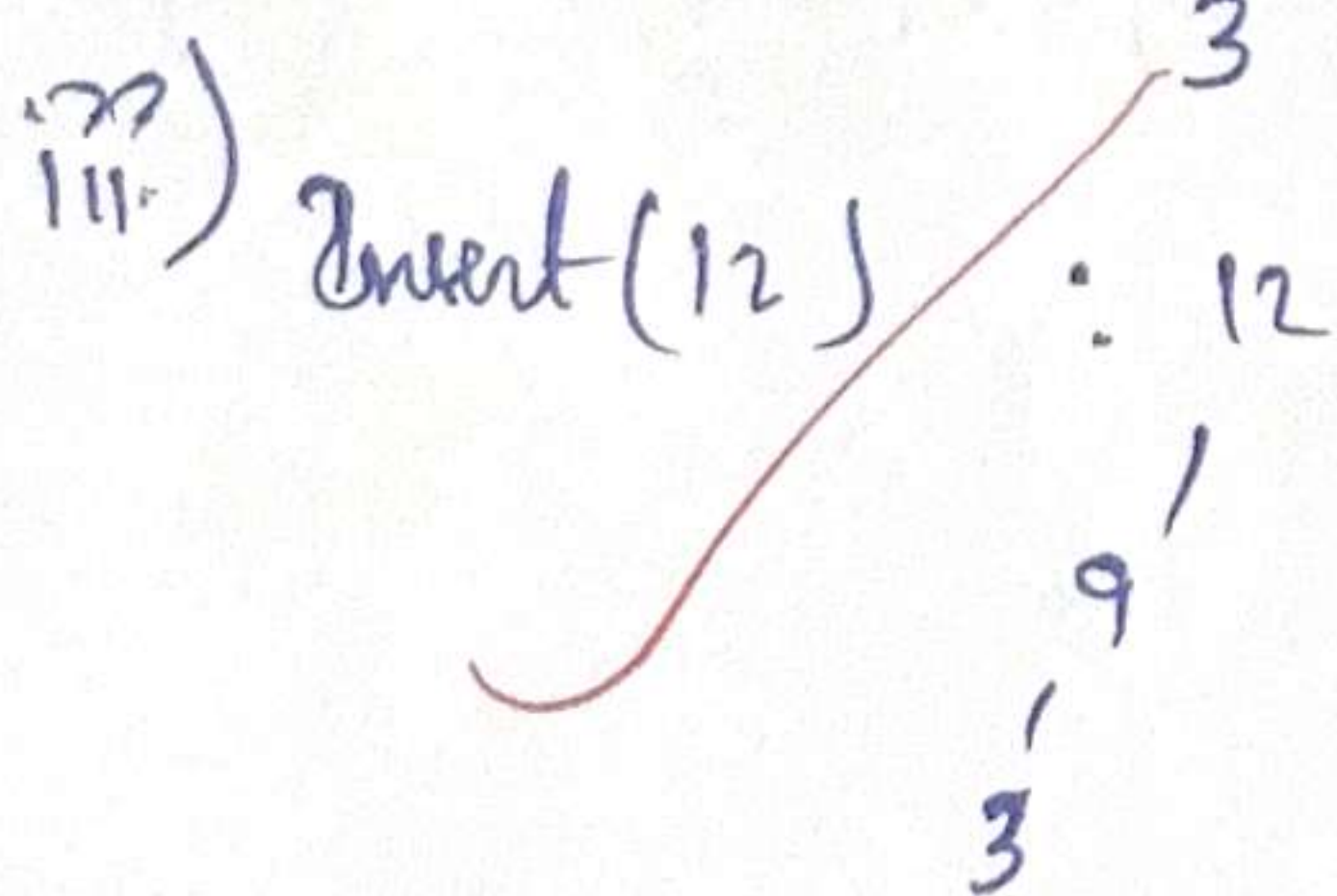


Array form: 45, 42, 35, 22, 23, 19, 21, 5, 20, 4

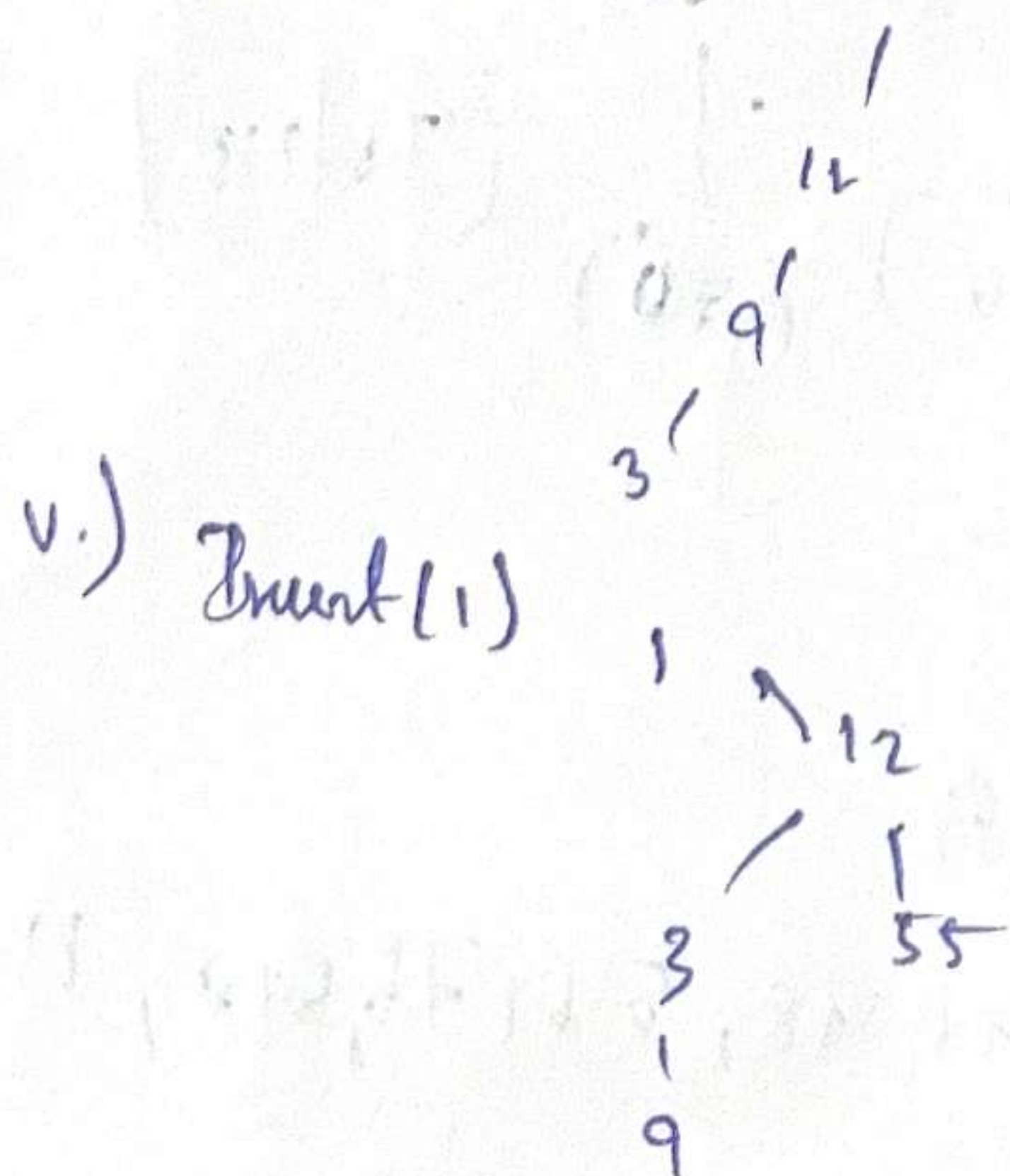
8.) B+ Tree operation:

i.) Insert (3) : 3

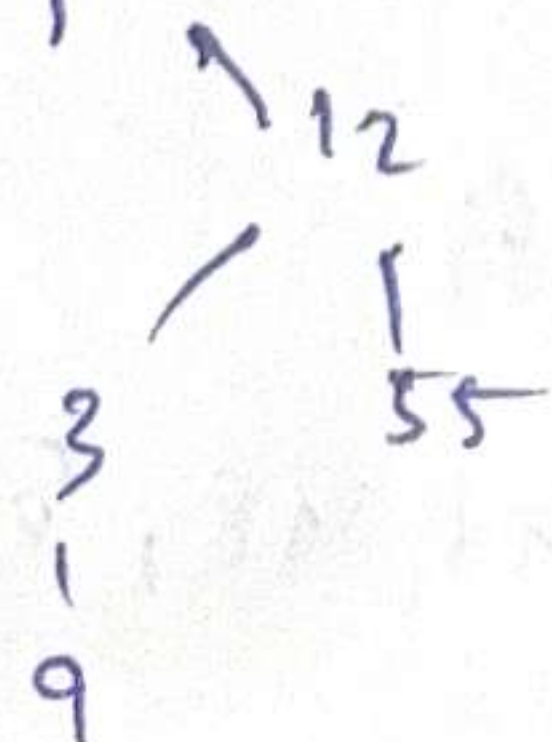
ii.) Insert (9) : 9



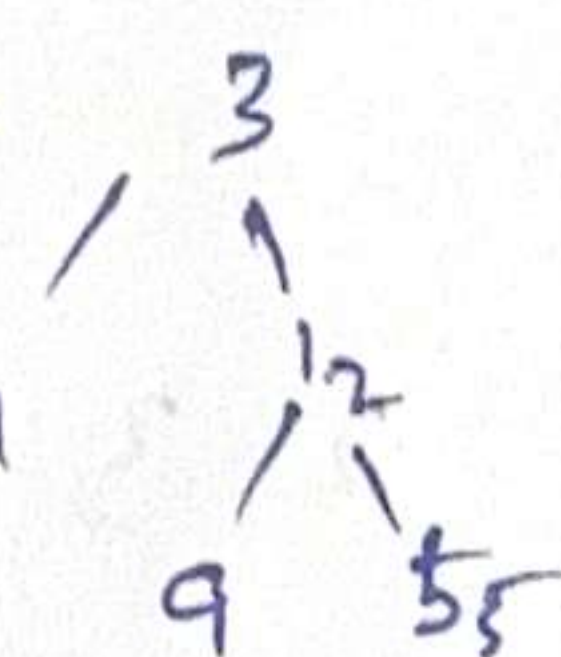
iv.) Insert (55) : 55



v.) Insert (1) :



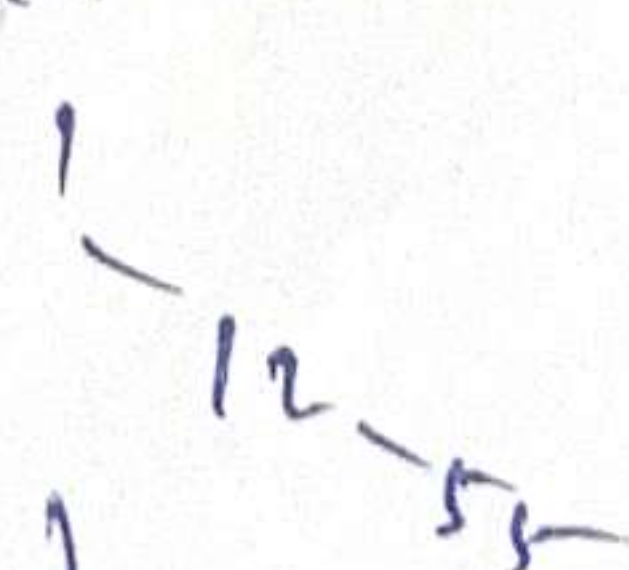
vi.) Insert (3) :



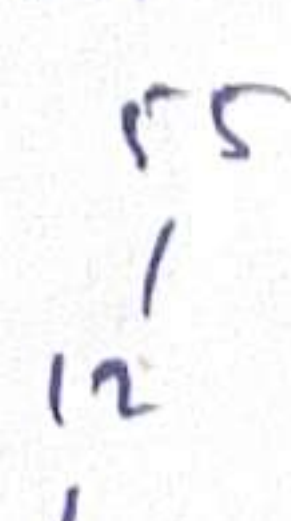
vii.) Delete (3) :



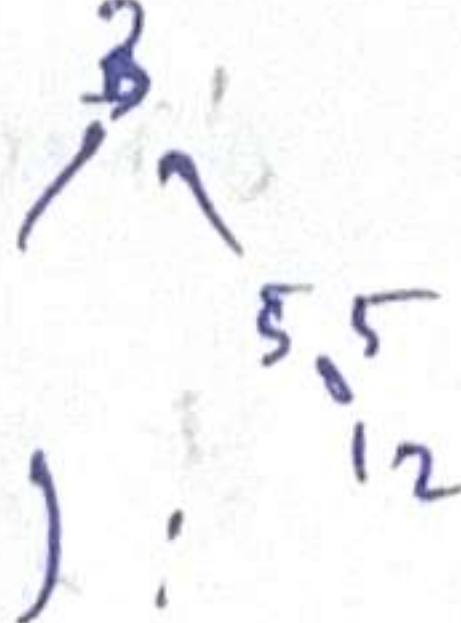
viii.) Delete (9) :



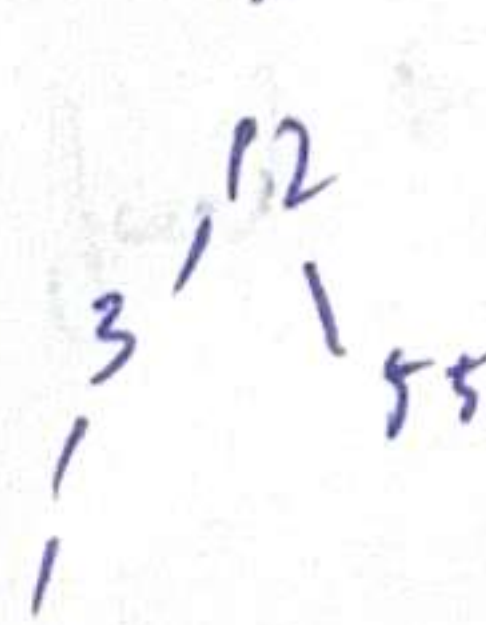
ix.) Delete (60) :



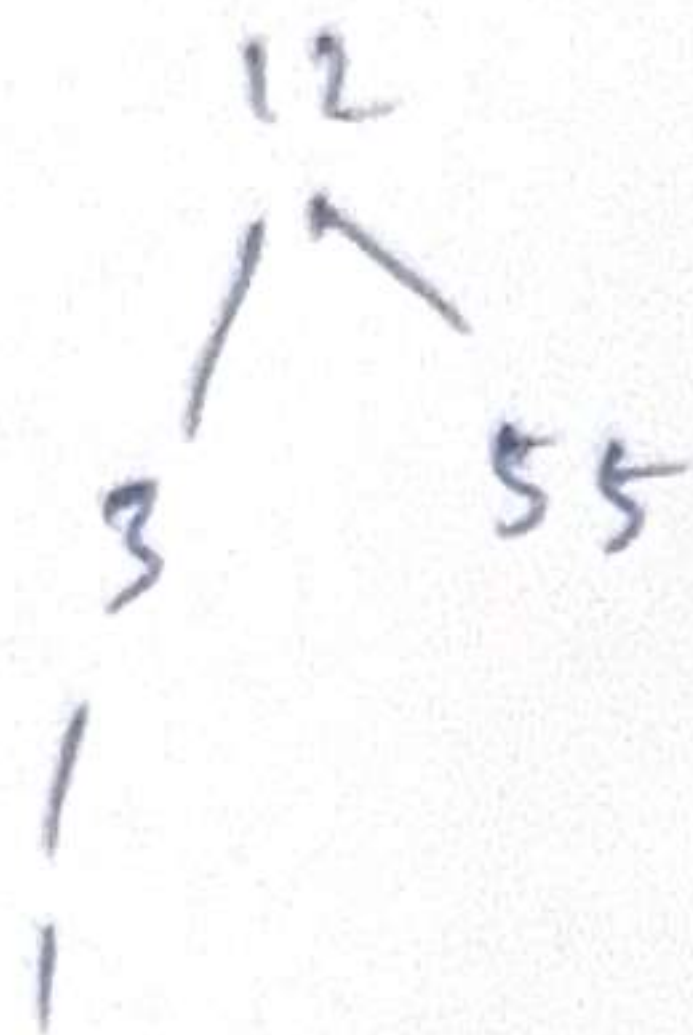
x.) Insert (3) :



xi.) Search (12) :



The final Merge Tree:



Radix Sort:

Given: 170, 45, 75, 90, 802, 24, 2, 66

i.) One Digit:

170, 90, 802, 2, 24, 45, 75, 66.

ii.) Two Digit:

802, 2, 24, 45, 66, 170, 75, 90.

iii.) Hundreds digit:

2, 24, 45, 66, 75, 90, 170, 802

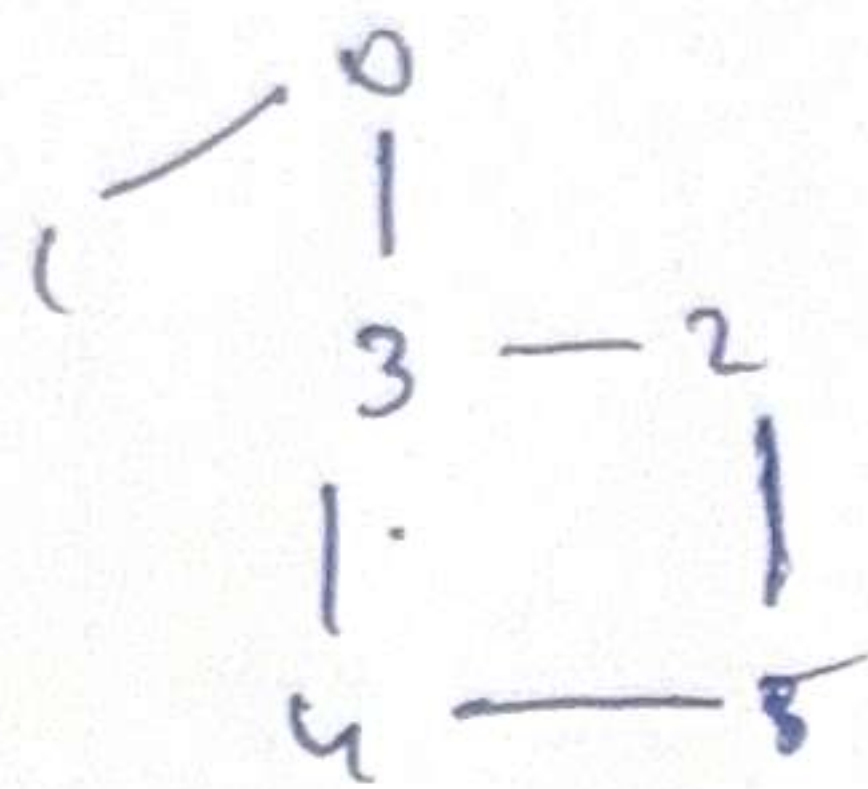
Final answer:

2, 24, 45, 66, 75, 90, 170, 802

Radix Sort: Sort numbers digit by digit starting from the rightmost digit.

10) BFS (Breadth First Search).

Given:



Start from 0

BFS Order:

0 → 1 → 3 → 2 → 4 → 5

Simple Steps:

\* Start: 0

\* Visit: 1, 3, 2.

\* Then: 4, 5

Final Breadth First Search

0 → 1 → 3 → 2 → 4 → 5

Urging